FOUNDATIONAL RESEARCH PROGRAM

for the Quantum Stellarator Architecture

Established Science, Engineering Building Blocks, and Concretization Roadmap

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This document inverts the thesis document's direction of exposition.

Where the thesis moves from architecture to foundations,
this program moves from established science to architecture.

Every entry is grounded in published, peer-reviewed work
or demonstrated engineering practice.
Speculative extensions are explicitly marked.

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Chapter 1

Program Overview

1.1 Purpose and Scope

The Quantum Stellarator thesis document describes an eight-layer architecture for topological quantum confinement, grounded in formal verification. That document moves from architecture downward to foundations. This research program inverts the direction: it begins with established science—peer-reviewed results, demonstrated fabrication techniques, commercially available materials, and operational experimental platforms—and maps those concrete building blocks upward onto the stellarator architecture.

The goal is to answer a specific question for each architectural layer: *what already exists, what needs to be built, and what is the gap between the two?*

1.2 Methodology: Technology Readiness Levels

Each building block is assessed using an adapted Technology Readiness Level (TRL) scale:

TRL	Level	Definition (adapted)
9	Operational system	Commercially deployed; off-the-shelf
8	System complete	Demonstrated in operational environment
7	Subsystem demo	Prototype in relevant environment
6	System/subsystem model	Demonstrated in relevant environment
5	Component validation	Validated in relevant environment
4	Lab validation	Component validated in laboratory
3	Proof of concept	Experimental proof of concept
2	Concept formulated	Technology concept formulated
1	Basic research	Basic principles observed

Table 1.1: Adapted TRL scale with Glassbox tier correspondence.

The TRL scale maps onto the Glassbox epistemic lattice: TRL 7–9 corresponds roughly to Proved/Demonstrated, TRL 4–6 to Demonstrated/Motivated, TRL 2–3 to Motivated/Conjectured, and TRL 1 to Speculative. This mapping is approximate but provides a shared vocabulary between the engineering and epistemic domains.

1.3 Document Structure

Each chapter addresses one layer pair of the stellarator architecture. Within each chapter:

1. **Established science** — peer-reviewed results forming the knowledge base
2. **Available materials and components** — what can be procured or fabricated today
3. **Demonstrated techniques** — fabrication, measurement, and characterization methods
4. **Key experimental precedents** — published experiments closest to what the architecture requires
5. **Gap analysis** — what the architecture demands vs. what currently exists
6. **Concretization tasks** — specific research tasks to close the gaps

Chapter 2

Layers 0–1: Superconducting Substrate and Quasicrystal Graphene

2.1 Established Science

2.1.1 Niobium Superconductivity

Niobium is the workhorse elemental superconductor with the highest critical temperature ($T_c \approx 9.3\,\mathrm{K}$) of any element. Its foundational science is mature:

- **BCS theory:** Niobium's superconducting properties are fully described by Bardeen–Cooper–Schrieffer theory. The energy gap $\Delta(0) \approx 1.55\,\mathrm{meV}$, the coherence length $\xi_0 \approx 38\,\mathrm{nm}$, and the London penetration depth $\lambda_L \approx 39\,\mathrm{nm}$ are precisely measured and tabulated.
- **Critical fields:** Type-II superconductor with $H_{c1} \approx 0.17\,\mathrm{T}$ and $H_{c2} \approx 0.4\,\mathrm{T}$ (bulk). Vortex physics well-characterized.
- **Surface science:** Nb_2O_5 native oxide formation, its impact on surface resistance, and mitigation strategies (nitrogen doping, electropolishing, low-temperature baking) extensively studied for SRF cavity applications.

TRL Assessment: 9 — Commercially deployed Bulk Nb, thin film Nb

Niobium superconductivity is among the most thoroughly characterized phenomena in condensed matter physics. Nb thin films are deposited routinely via magnetron sputtering, e-beam evaporation, and atomic layer deposition (ALD). SRF cavities operating at 1.8 K achieve quality factors exceeding 10^{11}.

2.1.2 Quasicrystals: Structure and Properties

- **Discovery and theory:** Dan Shechtman's 1982 observation of fivefold diffraction symmetry in Al–Mn alloys (Nobel Prize 2011). Penrose tilings (1974) provided the mathematical

framework before the physical discovery.

- **Phonon properties:** Quasicrystals exhibit anomalously low thermal conductivity — comparable to amorphous materials — despite possessing long-range order. This is the *phonon glass* property central to the PGTC mechanism. Measured in Al–Pd–Mn ($\kappa \approx 1$–$2\,\mathrm{W/m{\cdot}K}$, vs. $\sim$200 W/m·K for crystalline Al).
- **Electronic properties:** Pseudogap formation near the Fermi level, conductivity suppression relative to crystalline phases, but coherent diffraction demonstrates long-range electronic order.
- **Bott periodicity and K-theory:** The mathematical classification of topological phases in quasiperiodic systems uses noncommutative geometry (Bellissard, 1986). Gap labeling theorems connect spectral gaps to K-theory invariants of the hull of the tiling — this is the formal bridge between quasicrystal structure and topological classification.

TRL Assessment: 7 — Demonstrated in relevant environments Bulk quasicrystals, thin-film quasicrystals

Thermodynamically stable icosahedral quasicrystals (Al–Pd–Mn, Al–Cu–Fe) can be grown as single-grain samples. Thin-film quasicrystalline approximants have been deposited by MBE and sputtering. Phonon dispersion measured by inelastic neutron scattering.

2.1.3 Graphene: Synthesis and Patterning

- **CVD growth:** Chemical vapor deposition on Cu and Ni substrates produces large-area ($>1\,\mathrm{cm}^2$), predominantly monolayer graphene. Transfer techniques (PMMA-mediated, dry transfer) are routine.
- **Lithographic patterning:** Electron-beam lithography (EBL) achieves sub-10 nm feature resolution on graphene. Focused ion beam (FIB) milling provides complementary patterning. Nanosphere lithography enables periodic patterning at wafer scale.
- **Twisted bilayer graphene:** The "magic angle" ($\theta \approx 1.1$) discovery (Cao et al., 2018) demonstrated that moiré superlattices in graphene create flat electronic bands, correlated insulator states, and unconventional superconductivity. This is the closest existing experimental analogue to the quasiperiodic graphene concept in the stellarator architecture.
- **Graphene on Nb:** Graphene/Nb heterostructures have been fabricated for proximity-effect studies and Josephson junction research. The interface is compatible.

TRL Assessment: 8–9 — Graphene synthesis and patterning CVD graphene, EBL

patterning, transfer methods

Graphene growth and patterning are commercially available services. The specific challenge is patterning graphene into a *Penrose quasiperiodic* lattice, which requires aperiodic lithographic masks (TRL 3–4 for this specific application).

2.2 Available Materials and Components

Material	Form	TRL	Source/Supplier
Niobium	Bulk ingot, sheet, wire	9	CBMM, H.C. Starck, Plansee
Nb thin film	Sputtered/evaporated	9	Any PVD facility
Monolayer graphene	CVD on Cu foil	9	Graphenea, ACS Material, 2D Semiconductors
Graphene on Nb	Transferred heterostructure	5	Academic labs (proximity-effect studies)
Penrose-patterned graphene	EBL-patterned	3	Custom fabrication required
Quasicrystal substrates	Al–Pd–Mn single grain	4	Specialized crystal growth labs

Table 2.1: Materials inventory for layers 0–1.

2.3 Key Experimental Precedents

1. **Phonon transport in quasicrystals** (de Boissieu et al., 2007; Dugain et al., 1997): Inelastic neutron scattering measurements of phonon dispersion in i-Al–Pd–Mn showing anomalous thermal conductivity and critical mode dampening — direct evidence of the phonon glass property.

2. **Twisted bilayer graphene superlattices** (Cao et al., 2018; Yankowitz et al., 2019): Moiré superlattice engineering in graphene creating flat bands and correlated electron states. Demonstrates that geometric modification of graphene's lattice produces emergent electronic phases — the conceptual precursor to Penrose-patterned graphene.

3. **Graphene nanomesh fabrication** (Bai et al., 2010): Block-copolymer lithography creating periodic hole arrays in graphene with sub-20 nm features, opening a transport gap while maintaining high mobility. Demonstrates wafer-scale graphene patterning feasibility.

4. **Nb–graphene Josephson junctions** (Heersche et al., 2007; Calado et al., 2015): Supercurrent transport through graphene contacted by superconducting electrodes. Demon-

strates the Nb–graphene interface is functional for quantum transport.

5. **Quasiperiodic photonic crystals** (Vardeny et al., 2013): Penrose-tiled photonic structures showing omnidirectional band gaps absent in periodic crystals. While in photonics rather than phonons, this demonstrates that quasiperiodic patterning creates qualitatively different wave transport.

2.4 Gap Analysis

Architecture Requires	TRL	Current State	Gap
Nb superconducting substrate	9	Routine fabrication	None
Graphene monolayer synthesis	9	Commercial	None
Graphene transfer to Nb	5	Academic demo	Minor
Penrose-patterned graphene	3	Concept; periodic patterning exists	**Moderate**
Quasiperiodic graphene with verified phonon suppression	2	PGTC numerical prediction	**Major**
Coupled Nb + quasiperiodic graphene with verified topology	1–2	Theoretical	**Major**

Table 2.2: Gap analysis for layers 0–1.

2.5 Concretization Tasks

L01-1 Penrose mask design and fabrication. Design an EBL-compatible aperiodic mask based on a Penrose P3 tiling at the target length scale ($a \sim 10$–$50\,\mathrm{nm}$). Deliverable: GDS-II mask file validated against the PGTC's geometric parameters. *Estimated TRL advance: 3 → 4.*

L01-2 Quasiperiodic graphene patterning proof-of-concept. Fabricate Penrose-patterned graphene via EBL on a standard Si/SiO_2 substrate (not yet Nb). Characterize with Raman spectroscopy, STM/AFM for structural fidelity, and electron diffraction for aperiodic order confirmation. *Estimated TRL advance: 3 → 5.*

L01-3 Phonon transport measurement. Measure thermal conductivity of the quasiperiodic graphene sample using time-domain thermoreflectance (TDTR) or 3ω method. Compare directly against periodic graphene control sample. This is the **critical experimental test** of the PGTC prediction ($\kappa_{\mathrm{QP}}/\kappa_{\mathrm{ordered}} = 0.30$). *Estimated TRL advance: 2 → 4 (if PGTC prediction confirmed) or back to 1 (if refuted).*

L01-4 Graphene–Nb interface characterization. Transfer quasiperiodic graphene onto

Nb thin film. Characterize interface quality (contact resistance, proximity effect, critical current) at cryogenic temperatures. *Estimated TRL advance: 5 → 6.*

L01-5 Topological invariant measurement. Measure the winding number and edge state localization of the quasiperiodic graphene system via scanning tunneling spectroscopy (STS) at low temperature. Compare against PGTC predictions ($w = -1$, 99.7% edge localization). *Estimated TRL advance: 2 → 4.*

Chapter 3

Layers 2–3: Phonon Glass Thermal Management and NV Center Sensing

3.1 Established Science

3.1.1 The Phonon Glass Electron Crystal (PGEC) Concept

- **Origin:** Glen Slack (1995) proposed that ideal thermoelectrics would conduct electricity like a crystal and scatter phonons like a glass. The concept was realized in skutterudites ($CoSb_3$ with filler atoms), clathrates (type-I $Ba_8Ga_{16}Ge_{30}$), and half-Heusler compounds.
- **Mechanism:** Rattling guest atoms in oversized atomic cages scatter heat-carrying acoustic phonons without disrupting the charge-carrying electronic structure. Measured κ_{lattice} reductions of 50–90% relative to unfilled parent compounds.
- **Harrison/Keating models:** Harrison's universal scaling of the electron hopping integral $t(d) \propto d^{-2}$ (1980) and Keating's valence force field model giving phonon spring constants $k(d) \propto d^{-4}$ (1966) are standard references in semiconductor physics, routinely used in tight-binding and molecular dynamics calculations.

TRL Assessment: 6–7 — PGEC in bulk materials Skutterudites, clathrates, half-Heuslers

PGEC thermoelectrics are commercially produced (e.g., Marlow Industries). The concept is mature in 3D bulk. The stellarator's contribution is applying PGEC in a 2D quasiperiodic graphene geometry via the Harrison/Keating scaling asymmetry — that specific mechanism is TRL 2.

3.1.2 Nitrogen-Vacancy Centers in Diamond

NV centers are among the most mature quantum sensing platforms:

- **Atomic structure:** Substitutional nitrogen adjacent to a lattice vacancy in diamond. The NV^- charge state has spin-1 ground state with 2.87 GHz zero-field splitting.

- **Coherence:** $T_2 > 1\,\mathrm{ms}$ in isotopically purified ^{12}C diamond at room temperature (Balasubramanian et al., 2009). $T_2 > 1\,\mathrm{s}$ demonstrated at cryogenic temperatures.
- **Sensitivity:** DC magnetic field sensitivity $\sim 1\,\mathrm{nT}/\sqrt{\mathrm{Hz}}$ (single NV). AC sensitivity $\sim 10\,\mathrm{pT}/\sqrt{\mathrm{Hz}}$ with dynamical decoupling. Temperature sensitivity $\sim 100\,\mathrm{mK}/\sqrt{\mathrm{Hz}}$.
- **Fabrication:** Ion implantation (N^+ or N_2^+ at 5–30 keV) followed by annealing (800–1200 °C). Deterministic single-NV placement with ∼50 nm precision achieved via masked implantation.
- **Readout:** Optically detected magnetic resonance (ODMR). Photoluminescence at 637 nm (zero-phonon line) with broad phonon sideband to ∼800 nm. Confocal or widefield microscopy.

TRL Assessment: 7–8 — NV center sensing Single NV readout, NV ensemble magnetometry, NV thermometry

Commercial NV magnetometers exist (QZabre, Qnami). NV-based scanning probe microscopy achieves ∼10 nm spatial resolution. Integration with non-diamond substrates (e.g., NV in nanodiamonds on graphene) is TRL 4–5.

3.1.3 Diamond–Graphene Heterostructures

- **Interface:** Diamond and graphene share the carbon basis, enabling epitaxial relationships. Graphene on diamond (100) and (111) surfaces studied by several groups.
- **Nanodiamond on graphene:** Detonation nanodiamonds (∼5 nm) dispersed on graphene provide NV sensing capability in a 2D geometry. Demonstrated for magnetic imaging (Tetienne et al., 2017).
- **Diamane:** Fluorinated or hydrogenated bilayer graphene converts to a thin diamond-like (sp^3) film — a 2D diamond. Theoretically predicted to host NV centers (Bakharev et al., 2020).

TRL Assessment: 3–4 — Diamond–graphene integration Nanodiamond/graphene composites, diamane

Nanodiamond dispersal on graphene is routine. Deterministic NV placement in a nanodiamond-on-quasiperiodic-graphene architecture is TRL 2.

3.2 Available Materials and Components

Material	Form	TRL	Source
Electronic-grade diamond	CVD single crystal	9	Element Six, II-VI
NV diamond samples	Pre-characterized, single NV	8	QZabre, Qnami, academic
Detonation nanodiamonds	5–50 nm particles	9	Adamas Nano, Sigma-Aldrich
^{12}C isotopically pure diamond	CVD	6	Element Six (custom)
Skutterudite thermoelectrics	Bulk polycrystal	7	Marlow, commercial TE suppliers
Graphene with NV nanodiamonds	Composite	4	Academic fabrication

Table 3.1: Materials inventory for layers 2–3.

3.3 Key Experimental Precedents

1. **Thermal conductivity of quasicrystals** (Takeuchi et al., 2006): Systematic measurement across multiple quasicrystal families confirming $\kappa \approx 1$–$5\,\mathrm{W/m{\cdot}K}$ regardless of composition — evidence that the low κ is structural (quasiperiodicity) rather than compositional.

2. **NV thermometry at the nanoscale** (Kucsko et al., 2013): Single NV centers used to measure temperature with $\sim$10 mK precision and $\sim$200 nm spatial resolution inside living cells. Demonstrates the measurement capability the architecture needs for in-situ thermal monitoring.

3. **NV magnetometry on 2D materials** (Thiel et al., 2019): Scanning NV probe imaging of magnetism in atomically thin CrI_3. Demonstrates NV sensing of 2D material properties at the single-layer level.

4. **Phonon engineering in graphene superlattices** (Nika and Balandin, 2017): Review of phonon transport in graphene nanostructures showing that lateral confinement and patterning can reduce thermal conductivity by orders of magnitude.

5. **Anderson localization of phonons in disordered systems** (Hu et al., 2008): Experimental observation of phonon localization in a mesoscopic elastic network. Directly relevant to the PGTC's prediction of 12 localized phonon modes.

3.4 Gap Analysis

Architecture Requires	TRL	Current State	Gap
NV center fabrication and readout	8	Commercial products	None
PGEC in bulk thermoelectrics	7	Commercial products	None
NV sensing on graphene	4–5	Academic demonstrations	Minor
PGEC in 2D quasiperiodic graphene	2	PGTC numerical prediction	**Major**
Integrated NV + quasiperiodic graphene thermal management	1–2	Theoretical	**Major**

Table 3.2: Gap analysis for layers 2–3.

3.5 Concretization Tasks

L23-1 PGEC validation in 2D. This overlaps with L01-3 (phonon transport measurement). The critical addition: alongside thermal conductivity, measure electronic transport (Hall mobility, conductivity) on the same sample to confirm the PGEC property — low κ coexisting with reasonable σ. *TRL advance: 2 → 4.*

L23-2 Phonon mode spectroscopy. Use inelastic electron tunneling spectroscopy (IETS) or tip-enhanced Raman spectroscopy (TERS) to directly image localized phonon modes in the quasiperiodic graphene. Compare mode count and spatial distribution against the PGTC prediction (12 localized modes). *TRL advance: 2 → 4.*

L23-3 NV-on-quasiperiodic-graphene integration. Deposit nanodiamonds with pre-characterized NV centers onto the quasiperiodic graphene sample. Demonstrate ODMR readout of single NV centers through the graphene layer (graphene is optically transparent — this should work). *TRL advance: 4 → 5.*

L23-4 In-situ NV thermometry during transport. Use the integrated NV sensors to map the temperature field across the quasiperiodic graphene during electrical transport measurements. This simultaneously validates the PGEC property and the NV sensing integration. *TRL advance: 2 → 5 (if successful).*

L23-5 Spectral gap measurement. Measure the electronic spectral gap of the quasiperiodic graphene via STS. Compare against the PGTC prediction (58.8× spectral gap ratio). This is the experimental test of topological protection robustness. *TRL advance: 2 → 4.*

Chapter 4

Layers 4–5: Superfluid Medium and Topological Confinement

Epistemic status: This chapter has the widest gap between established science (which is deep) and the architecture's specific requirements (which are speculative). The foundational physics is Nobel Prize–caliber; the application to this geometry is largely uncharted.

4.1 Established Science

4.1.1 Helium Superfluidity

- **He-4:** Bose–Einstein condensation below $T_\lambda = 2.17\,\mathrm{K}$. Two-fluid model (Landau, Tisza). Quantized vortices with circulation quantum $\kappa = h/m_4 \approx 9.97 \times 10^{-8}\,\mathrm{m}^2/\mathrm{s}$. Fountain effect, second sound, roton minimum in dispersion.
- **He-3:** Fermionic superfluid below $T_c \approx 1$–$2.5\,\mathrm{mK}$ (depending on pressure). Cooper pairing in p-wave, spin-triplet channel. Multiple superfluid phases (A, B, A_1, planar) with distinct order parameter symmetries. ^{3}He-B has a fully gapped quasiparticle spectrum with topological surface states — a topological superfluid.
- **^{3}He-B as topological superfluid:** Class DIII in the Altland–Zirnbauer classification. Supports Majorana fermion surface bound states (predicted: Volovik, 2003; indirect evidence: Murakawa et al., 2009). The connection to topological classification is the same mathematical framework used in the GoS theorems.

TRL Assessment: 8–9 — Helium superfluid production and handling Cryostats, dilution refrigerators, superfluid instrumentation

He-4 superfluidity is routinely produced in any cryogenics lab. He-3 superfluidity requires millikelvin temperatures (dilution refrigerators or nuclear demagnetization cryostats) but is well within established experimental capability. Commercial dilution refrigerators (BlueFors, Oxford, Leiden Cryogenics) reach 5–10 mK routinely.

4.1.2 Analog Gravity in Superfluids

- **Unruh's proposal (1981):** Sound waves in a moving fluid obey the same wave equation as a scalar field in curved spacetime, with the acoustic metric determined by the fluid velocity profile. A sonic horizon (where flow velocity equals sound speed) creates an analog black hole.
- **Volovik's program:** The ^{3}He vacuum as a model for the quantum vacuum of particle physics. Emergent Lorentz invariance, emergent gauge fields, and analog Hawking radiation from topological defects in the superfluid order parameter.
- **Experimental demonstrations:** Weinfurtner et al. (2011) observed stimulated Hawking emission in a water wave analog. Švančara et al. (2024) reported thermal Hawking radiation analogs in superfluid ^{4}He. Jeff Steinhauer (2016, 2019) demonstrated analog Hawking radiation in BEC systems.

TRL Assessment: 4–5 — Analog gravity experiments BEC analogs, water wave analogs, superfluid helium analogs

Analog Hawking radiation has been observed in multiple platforms. The connection to actual quantum gravity remains theoretical. The stellarator architecture's use of analog gravity concepts (SVT) is TRL 1–2 at best.

4.1.3 Topological Defects in Superfluids

- **Quantized vortices:** Both ^{4}He and ^{3}He support quantized vortex lines. In ^{3}He, vortices carry additional topological structure due to the multicomponent order parameter.
- **Vortex knots:** Theoretical predictions of knotted vortex configurations in superfluids (Kleckner and Irvine, 2013, demonstrated knotted vortex lines in water). Superfluid vortex reconnection dynamics studied experimentally and numerically.
- **Trefoil vortex knots:** Kleckner and Irvine (2013) created trefoil-knotted vortex lines using 3D-printed hydrofoils in water. The trefoil is topologically stable until reconnection events unknot it — lifetime depends on fluid properties and geometry.

TRL Assessment: 3–4 — Knotted vortex lines in classical fluids Water vortex knots, numerical simulation of superfluid knots

Trefoil vortex knots have been created in classical fluids. Superfluid vortex knots have been studied numerically (Barenghi group). Creating and stabilizing a trefoil vortex knot in superfluid helium is TRL 2–3. Using a physical trefoil wire to template the vortex topology is TRL 1–2.

4.2 Available Materials and Components

Material	Form	TRL	Source
^{4}He	Liquid/superfluid, >99.999%	9	Air Liquide, Linde, Praxair
^{3}He	Gas/liquid, isotopically enriched	8	Limited supply; monitored
Dilution refrigerators	Base temp. 5–10 mK	9	BlueFors, Oxford, Leiden
Ni wire	Various gauges	9	Commercial suppliers
Ni-62 isotopically enriched wire	Custom	2	Isotope separation required
Trefoil-wound Ni-62 wire	Custom geometry	1	Not yet fabricated

Table 4.1: Materials inventory for layers 4–5.

4.3 Gap Analysis

Architecture Requires	TRL	Current State	Gap
Superfluid ^{4}He production	9	Routine	None
Superfluid ^{3}He production	8	Millikelvin cryogenics	None
Analog gravity experiments	4–5	Multiple platforms demonstrated	Minor
Knotted vortex lines in superfluids	2–3	Numerical; classical fluid demos	**Moderate**
Ni-62 trefoil confinement wire	1–2	Concept only	**Major**
SVT/ER=EPR experimental signatures	1	Theoretical	**Fundamental**

Table 4.2: Gap analysis for layers 4–5.

4.4 Concretization Tasks

L45-1 Trefoil vortex stability analysis. Numerical simulation (Gross–Pitaevskii equation for ^{4}He; Ginzburg–Landau for ^{3}He) of trefoil-knotted vortex lines in a confined geometry matching the stellarator's dimensions. Key question: what is the lifetime of the trefoil knot before reconnection unknots it? What confinement geometry maximizes stability? *TRL advance: 2 → 3.*

L45-2 Trefoil wire fabrication feasibility. Assess whether a trefoil knot can be wound from fine-gauge wire ($\sim$100 μm diameter) with sufficient precision for the required

magnetic field geometry. Start with Ni (natural isotope abundance) before considering Ni-62 enrichment. *TRL advance: 1 → 3.*

L45-3 Superfluid helium in confined geometries. Experimentally study superfluid ^{4}He behavior in a toroidal channel with nontrivial topology (a torus knot rather than a simple torus). Measure persistent current stability, vortex nucleation thresholds, and thermal transport. This does *not* require the full stellarator geometry — a simplified trefoil channel suffices. *TRL advance: 2 → 4.*

L45-4 ^{3}He-B surface states in confined geometry. Measure quasiparticle transport in ^{3}He-B confined in a channel with controlled surface topology. Look for signatures of Majorana surface bound states (anomalous thermal conductance, spin current). Connect to the Altland–Zirnbauer DIII classification verified in GoS. *TRL advance: 3 → 5.*

L45-5 Ni-62 isotope enrichment cost/feasibility study. Determine whether Ni-62 enrichment (via gas centrifuge or electromagnetic separation) is economically feasible at the required quantity (∼grams to ∼kilograms). The choice of Ni-62 is motivated by its nuclear spin properties; if enrichment is prohibitive, identify alternative isotopes or natural-abundance alternatives. *TRL advance: 1 → 2.*

Chapter 5

Layers 6–7: Bismuth Medium and OAM Laser Control

5.1 Established Science

5.1.1 Bismuth: Topological Semimetal

- **Electronic structure:** Bismuth has been studied since the 1920s (de Haas–van Alphen effect). Extremely small Fermi surface, high carrier mobility ($\mu \sim 10^6\,\mathrm{cm}^2/\mathrm{V{\cdot}s}$ at 4 K), low carrier density ($\sim 3 \times 10^{17}\,\mathrm{cm}^{-3}$), and long mean free path (∼mm at low temperature).
- **Topological properties:** Bi is a topological semimetal with strong spin–orbit coupling ($\Delta_{\mathrm{SOC}} \sim 1.5\,\mathrm{eV}$). Surface states confirmed by ARPES (Ast and Höchst, 2001). Bi_2Se_3 and Bi_2Te_3 are prototypical topological insulators; pure Bi is the precursor material.
- **Nuclear spin:** ^{209}Bi is 100% naturally abundant, spin-9/2, with the largest known nuclear magnetic moment among stable isotopes. This makes Bi an interesting medium for nuclear spin-based quantum protocols.
- **Diamagnetism:** Bismuth is the most strongly diamagnetic element — relevant for magnetic field shaping in the stellarator geometry.

TRL Assessment: 8–9 — Bismuth materials and characterization Bulk Bi crystals, Bi thin films, Bi nanowires

High-purity Bi single crystals are commercially available. Thin film deposition (MBE, sputtering) is routine. Bi nanowires fabricated by electrodeposition in anodic alumina templates. Topological surface states well-characterized.

5.1.2 Orbital Angular Momentum (OAM) of Light

- **Discovery:** Allen et al. (1992) showed that Laguerre–Gaussian laser modes carry quantized OAM of $\ell\hbar$ per photon, where ℓ is an integer (the topological charge). This is distinct from spin angular momentum (polarization).

- **Generation:** Spiral phase plates, spatial light modulators (SLMs), q-plates, fork holograms, and cylindrical lens mode converters all produce OAM beams routinely. Single-photon OAM states prepared and measured for quantum information.
- **High-dimensional encoding:** OAM provides a theoretically unbounded discrete Hilbert space ($\ell = 0, \pm 1, \pm 2, \ldots$), enabling high-dimensional quantum key distribution (Mirhosseini et al., 2015) and dense information encoding.
- **OAM in matter:** Electron vortex beams carrying OAM demonstrated in electron microscopy (Uchida and Tonomura, 2010; Verbeeck et al., 2010). Phonon OAM in crystals proposed theoretically (Zhang and Niu, 2015).

TRL Assessment: 7–8 — OAM beam generation and detection SLMs, q-plates, spiral phase plates, single-photon OAM

OAM beams are commercially generated (Hamamatsu SLMs, Thorlabs optics). Single-photon OAM encoding demonstrated for QKD. Application to quantum state control in a topological material (the stellarator's use case) is TRL 2–3.

5.1.3 ZX Calculus

- **Formalism:** Coecke and Duncan (2008, 2011) introduced ZX calculus as a graphical language for quantum computation based on complementary observables. It is complete for stabilizer quantum mechanics and approximately universal.
- **Verification:** ZX calculus has been used for circuit optimization (PyZX), quantum error correction analysis, and measurement-based quantum computation verification. It provides a natural bridge between quantum circuits and category theory.
- **Software:** PyZX (open-source Python library) implements ZX calculus simplification rules for quantum circuit optimization.

TRL Assessment: 6–7 — ZX calculus for circuit verification PyZX, theoretical framework

ZX calculus is mature as a mathematical framework. Its application to physical OAM-based quantum control (rather than abstract circuit optimization) is TRL 2.

5.2 Concretization Tasks

L67-1 Bi thin film topological characterization at cryogenic temperatures. Fabricate high-quality Bi(111) thin films on a suitable substrate. Characterize topological surface states via ARPES and magneto-transport at temperatures compatible with the Nb superconducting layer ($T < 9\,\mathrm{K}$). Measure spin–orbit coupling strength and surface state dispersion. *TRL advance: 8 → 8 (confirmation in stellarator-relevant conditions).*

L67-2 OAM beam interaction with Bi surface states. Explore whether OAM-carrying photons couple differently to topological surface states than conventional Gaussian beams (polarization-only). Theoretical prediction: the OAM degree of freedom may couple to the orbital texture of the surface states. *TRL advance: 1 → 3.*

L67-3 ZX calculus compilation for OAM control sequences. Implement a ZX calculus compiler that targets OAM-encoded quantum states instead of standard qubit gates. The compiler would take a quantum circuit specified in ZX calculus and produce a sequence of SLM phase patterns implementing the desired operation on OAM modes. *TRL advance: 2 → 4.*

L67-4 Bi nuclear spin coherence measurement. Measure nuclear spin coherence times of ^{209}Bi in single-crystal Bi at millikelvin temperatures. The spin-9/2 system provides a 10-dimensional Hilbert space per nucleus — assess whether this is usable for quantum information storage. *TRL advance: 3 → 5.*

Chapter 6

Stellarator Integration Topos

6.1 Established Science

6.1.1 Stellarator Plasma Confinement

- **History and operation:** Lyman Spitzer proposed the stellarator concept in 1951. Wendelstein 7-X (W7-X) at Max Planck Institute for Plasma Physics began operating in 2015 and is the world's largest stellarator. It has achieved plasma confinement times of 8+ minutes and demonstrated the feasibility of steady-state magnetic confinement without plasma current.
- **Coil geometry:** Stellarator magnetic fields are produced entirely by external coils (unlike tokamaks, which require a plasma current). W7-X uses 50 non-planar superconducting NbTi coils and 20 planar coils. The coil shapes are computationally optimized to produce a specific magnetic field geometry.
- **Optimization:** Modern stellarators are designed by numerical optimization of the magnetic field geometry to minimize neoclassical transport. This is a solved computational problem — the tools exist (VMEC, STELLOPT, SIMSOPT).

TRL Assessment: 7–8 — Stellarator design and operation W7-X, HSX, computational design tools

Stellarator design is a mature engineering discipline. The Quantum Stellarator borrows the geometric concept (external-coil magnetic confinement with nontrivial topology) but at a completely different scale and for a different purpose (quantum state confinement rather than plasma confinement).

6.1.2 Trefoil Knot Topology in Physics

- **Knot invariants:** The trefoil is the simplest nontrivial knot (crossing number 3). Its Jones polynomial, Alexander polynomial, and knot group ($\langle a, b \mid a^2 = b^3 \rangle$, the braid group B_3) are completely characterized.

- **Physical realizations:** Knotted electromagnetic fields (Irvine, 2010), knotted vortex lines in fluids (Kleckner and Irvine, 2013), knotted dislocations in liquid crystals (Tkalec et al., 2011), and knotted DNA (Arsuaga et al., 2005) all demonstrate that trefoil topology can be physically instantiated.
- **Topological protection:** Knot invariants are topologically protected — they cannot change under continuous deformation. This is the fundamental reason for choosing a knotted geometry: a trefoil knot cannot be unknotted by any smooth perturbation, providing inherent stability to the confinement geometry.

TRL Assessment: 3–4 — Trefoil topology in physical systems Knotted vortices, knotted optical fields, knotted defects

Trefoil topology has been created in multiple physical systems. None of these demonstrations involve quantum confinement. The use of trefoil topology for quantum state confinement is TRL 1–2.

6.2 Concretization Tasks

ST-1 Magnetic field simulation for trefoil geometry. Use stellarator design codes (SIMSOPT or similar) to compute the magnetic field produced by a trefoil-wound coil at the scale of the quantum stellarator (∼cm rather than ∼m). Assess field homogeneity, gradient control, and compatibility with Nb superconducting operation. *TRL advance: 2 → 3.*

ST-2 Layer interface compatibility matrix. Systematically assess the physical compatibility of adjacent layers: thermal expansion mismatch, chemical reactivity, electrical contact, and cryogenic cycling tolerance for every interface (Nb/graphene, graphene/diamond, diamond/He, He/Bi). Produce a materials compatibility database. *TRL advance: 2 → 3.*

ST-3 Topos-theoretic composition verification. Formally verify in Lean 4 or Cubical Agda that the mathematical structures assigned to each layer compose coherently in the stellarator topos. This is a pure formal verification task, not an experimental one, but it is the integration test for the entire theoretical architecture. *TRL advance (formal): 2 → 4.*

ST-4 Scale-model magnetic confinement. Fabricate a simple trefoil-wound Nb coil (natural Nb, not Ni-62) and characterize its magnetic field at 4 K. This tests the most basic engineering question: can the geometry be built and does it produce the

expected field? *TRL advance: 1 → 4.*

Chapter 7

Formal Verification Infrastructure

7.1 Established Tools and Platforms

The formal verification substrate is itself a technology stack with its own readiness levels:

Tool/Platform	TRL	Status
Lean 4	9	Stable release; Mathlib ecosystem; active development
Cubical Agda	8	Stable; 1Lab library; smaller community than Lean
GoS framework (this project)	5	144 theorems; zero Mathlib dependencies; audited
TopologicalBridge	3	Porting in progress
Omnivalence Postulate	2	Conjectured; no formal proof
PyZX	6	Open source; circuit optimization
Azure Quantum / Materials	7	Cloud platform; DFT and molecular simulation
CUDA-Q / NVIDIA	7	GPU-accelerated quantum circuit simulation

Table 7.1: Formal verification and simulation infrastructure TRL.

7.2 Concretization Tasks

FV-1 GoS theorem dependency graph. Produce a complete dependency graph of all 144 GoS theorems, showing which results each theorem depends on (noting that all are zero-dependency by design, but the *conceptual* dependencies matter for understanding the proof architecture). Publish alongside the repository. *TRL advance: 5 → 6.*

FV-2 CLHoTT sorry elimination. Address the six remaining `sorry` statements in `CLHoTT.lean`. These are Float ring law axioms. Options: replace Float with a rational arithmetic library, axiomatize the ring laws explicitly with documented assumptions, or port to a representation where the ring laws hold definitionally. *TRL advance: 5 → 6.*

FV-3 TopologicalBridge completion. Complete the port of topological theorems from Lean 4 to Cubical Agda. Priority: the winding number computation and edge state localization theorems, which are most naturally expressed in HoTT. *TRL advance: 3 → 5.*

FV-4 Experimental prediction extraction. Write a Lean 4 program that extracts concrete numerical predictions from the GoS theorems in a format directly comparable to experimental measurements. The bridge between "theorem: winding number $= -1$" and "experiment: measure X, expect Y" must be explicit and auditable. *TRL advance: 2 → 4.*

Chapter 8

Phased Research Roadmap

The concretization tasks from all chapters are organized into a phased execution plan. The phases are sequenced by dependency (later phases require results from earlier phases) and by risk (earlier phases test the highest-risk assumptions first).

8.1 Phase 1: Foundation Validation (Months 1–6)

Epistemic status: Goal: Experimentally test the PGTC's core predictions. This is the make-or-break phase.

Task	Description	Priority
L01-1	Penrose mask design and GDS-II file	Critical path
L01-2	Quasiperiodic graphene patterning PoC	Critical path
L01-3	Phonon transport measurement (PGTC test)	**Gating**
L23-1	PGEC validation (electronic + thermal transport)	**Gating**
FV-1	GoS dependency graph publication	Parallel
FV-2	CLHoTT sorry elimination	Parallel

Table 8.1: Phase 1 tasks. **Gating** tasks determine whether the program continues.

Go/no-go criterion: If L01-3 and L23-1 confirm phonon suppression within a factor of 2 of the PGTC prediction ($0.15 < \kappa_{\mathrm{QP}}/\kappa_{\mathrm{ordered}} < 0.60$), proceed to Phase 2. If phonon suppression is not observed, return to theory and revise the PGTC model before further experimental investment.

8.2 Phase 2: Topological Characterization (Months 4–12)

Epistemic status: Goal: Measure topological invariants and integrate NV sensing. Overlaps with Phase 1.

Task	Description	Priority
L01-4	Graphene–Nb interface characterization	High
L01-5	Winding number and edge state measurement	Critical path
L23-2	Phonon mode spectroscopy	High
L23-3	NV-on-quasiperiodic-graphene integration	High
L23-5	Spectral gap measurement	High
FV-3	TopologicalBridge completion	Parallel
FV-4	Experimental prediction extraction program	Parallel

Table 8.2: Phase 2 tasks.

8.3 Phase 3: Layer Integration and Extended Physics (Months 9–18)

Epistemic status: Goal: Begin integrating adjacent layers and explore the superfluid/control layers.

Task	Description	Priority
L23-4	In-situ NV thermometry during transport	High
L45-1	Trefoil vortex stability (numerical)	Medium
L45-2	Trefoil wire fabrication feasibility	Medium
L67-1	Bi thin film cryo characterization	Medium
L67-3	ZX calculus compiler for OAM	Medium
ST-1	Trefoil magnetic field simulation	Medium
ST-2	Layer interface compatibility matrix	High

Table 8.3: Phase 3 tasks.

8.4 Phase 4: Stellarator-Scale Integration (Months 15–24)

Epistemic status: Goal: Build toward integrated prototypes. Conditional on Phase 1–3 results.

Task	Description	Priority
L45-3	Superfluid He in confined trefoil geometry	High
L45-4	^{3}He-B surface states in controlled topology	High
L45-5	Ni-62 enrichment feasibility study	Medium
L67-2	OAM beam–Bi surface state interaction	High
L67-4	^{209}Bi nuclear spin coherence	Medium
ST-3	Topos composition formal verification	High
ST-4	Scale-model Nb trefoil coil	High

Table 8.4: Phase 4 tasks.

Chapter 9

Resource Requirements

9.1 Facility Access

The research program requires access to the following categories of experimental infrastructure:

Capability	Required For	Access Model
Electron-beam lithography	L01-1, L01-2	University cleanroom or NNCI[1]
CVD graphene growth	L01-2	University lab or commercial
Cryogenic transport	L01-3, L01-4, L23-1	PPMS or custom cryostat
TDTR / 3ω thermal measurement	L01-3, L23-1	Specialized thermal lab
Low-temperature STM/STS	L01-5, L23-2, L23-5	University surface science
Confocal microscopy (NV readout)	L23-3, L23-4	University quantum optics
Millikelvin cryogenics	L45-3, L45-4	Dilution refrigerator lab
MBE / sputtering (Bi films)	L67-1	University thin-film lab
Spatial light modulator optics	L67-2, L67-3	University photonics
NbTi/Nb coil winding	ST-4	Superconducting magnet shop

Table 9.1: Facility requirements by task.

9.2 Computational Resources

- **Lean 4 / Cubical Agda:** Desktop-class hardware suffices for all formal verification tasks.
- **Azure Quantum:** Cloud-based; subscription or grant access. Required for DFT calculations (phonon dispersion, electronic band structure) in L01-3, L23-1, L23-2.
- **CUDA-Q:** Requires NVIDIA GPU cluster access. Required for quantum circuit simulation in L67-3 and stellarator integration tasks.
- **Stellarator optimization codes:** SIMSOPT (open source, Python); runs on standard HPC. Required for ST-1.

[1] National Nanotechnology Coordinated Infrastructure — a network of university-based nanofabrication facilities with user-access programs.

9.3 Estimated Budget by Phase

Phase	Primary Costs	Estimate
Phase 1	EBL time, CVD graphene, cryo measurement, cloud compute	\$80K–150K
Phase 2	STM/STS time, NV diamond materials, confocal setup	\$100K–200K
Phase 3	DR time, Bi MBE, SLM optics, numerical simulation	\$150K–300K
Phase 4	Isotope enrichment, custom cryostat, coil fabrication	\$200K–500K
Total		**\$530K–\$1.15M**

Table 9.2: Estimated budget ranges by phase. Assumes university facility access rates, not commercial.

These estimates assume university user-facility rates and do not include personnel costs. A single-PI model with 1–2 graduate students or postdocs would add \$150K–300K/year in personnel. Total program cost over 24 months: approximately \$0.8M–\$1.75M, competitive with a standard NSF CAREER or DOE Early Career award.

Chapter 10

Risk Assessment

10.1 Risk Matrix

Risk	Likelihood	Impact	Mitigation
PGTC phonon suppression not observed experimentally	Medium	**Fatal**	Phase 1 go/no-go gate. Revise model before further investment.
Penrose-patterned graphene structurally unstable	Low	High	Fall back to Fibonacci-chain 1D structures for initial tests
NV centers incompatible with graphene substrate	Low	Medium	Use scanning NV probe (non-contact) instead of integrated NV
Trefoil vortex knot lifetime too short	Medium	Medium	Explore pinning mechanisms; modified geometry
Ni-62 enrichment prohibitively expensive	High	Low	Use natural Ni; Ni-62 optimization is non-essential
SVT/ER=EPR connections experimentally inaccessible	High	Low	Layers 4–5 are not load-bearing for the verified core
Omnivalence Postulate unprovable	Medium	Low	GoS and Agda operate independently; bridge is optional
GoS sorry count increases during development	Low	Medium	Strict sorry accounting; CI/CD checks on every commit

Table 10.1: Risk matrix with Glassbox-colored likelihood assessment.

10.2 Critical Path

The single critical path through the entire program is:

L01-1 (mask design) → **L01-2** (patterning) → **L01-3** (phonon measurement) → **go/no-go**

Everything else can proceed in parallel or is contingent on this result. If the phonon suppression measurement confirms the PGTC prediction, the program has an experimentally validated foundation. If it does not, the theoretical model must be revised before the remaining $400K–$1M of experimental work is justified.

This is, by design, the same logic encoded in the project navigation flowchart's outreach branch: build the foundation before building on it.

End of Document

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Foundational Research Program for the Quantum Stellarator Architecture